

Talazoparib in Human Breast Cancer

Talazoparib (ChEMBL: CHEMBL3137320 ¹; DrugBank: DB11760 ²; brand name **Talzenna** ³) is a potent small-molecule inhibitor of poly(ADP-ribose) polymerases (PARP-1/2). It was FDA-approved in October 2018 for treatment of **germline BRCA1/2-mutated, HER2-negative** advanced or metastatic breast cancer ⁴ ⁵. Key synonyms include BMN-673, LT-00673, and MDV-3800 ⁶. In breast cancer, talazoparib exploits **homologous recombination (HR) repair deficiency**; it selectively kills HR-deficient tumor cells (a concept known as synthetic lethality) ⁷ ⁸.

Talazoparib's **mechanism of action** involves dual inhibition and trapping of PARP enzymes. It binds with high affinity to PARP1/2, blocking base-excision repair of single-strand DNA breaks ⁷. Unrepaired single-strand breaks accumulate and convert into toxic double-strand breaks (DSBs) during replication. In normal cells, DSBs are repaired by the BRCA1/2-dependent HR pathway ⁷; but in BRCA1/2-deficient cancer cells this repair is defective, leading to **genomic instability and apoptosis** (synthetic lethality) ⁹. Moreover, talazoparib is a highly potent "PARP-trapping" inhibitor: it stabilizes PARP-DNA complexes at sites of damage, causing replication fork collapse and further DNA damage ⁹ ¹⁰. This dual action – catalytic inhibition and PARP trapping – markedly increases talazoparib's cytotoxicity in HRD tumors. For example, talazoparib markedly induces apoptosis in BRCA1/2-mutant breast cancer cells ⁹ ¹¹, whereas restoration of HR repair (via secondary BRCA mutations or other mechanisms) abrogates its effect ¹² ¹¹.

Clinically, talazoparib is **indicated** for patients with deleterious or suspected deleterious **germline BRCA1/2 mutations in HER2-negative breast cancer** ¹³. The pivotal phase III EMBRACA trial showed that talazoparib significantly prolonged progression-free survival compared to physician's choice chemotherapy in gBRCA-mutant metastatic breast cancer ¹⁴. Although overall survival benefits were modest, quality-of-life favored talazoparib. Guidelines now recommend germline BRCA testing in all recurrent/metastatic breast cancer patients ⁵ to identify candidates for PARP inhibitors (including talazoparib) as standard therapy. Talazoparib is typically used in **HER2-negative** disease (both triple-negative and hormone-receptor positive), reflecting its trial populations ¹⁵ ¹³. In practice, TNBC (often basal-like) and luminal subtypes with BRCA mutations show similar sensitivity: an analysis of the ABRAZO trial found comparable talazoparib efficacy in BRCA1-mutant (typically basal/TNBC) and BRCA2-mutant (often luminal) tumors ¹⁶. Notably, the most common co-mutations were *TP53* in BRCA1 tumors and *PIK3CA* in BRCA2 tumors ¹⁶, but these did not significantly alter talazoparib response ¹⁶.

Resistance and Sensitivity Pathways

Several molecular pathways modulate talazoparib **sensitivity** or **resistance**. Key examples include:

- **Homologous Recombination Repair (HRR) [R-HSA-5685942]:** *Downregulation* (HR deficiency) strongly sensitizes to talazoparib (synthetic lethality) ⁸ ⁷. Patients with BRCA1/2 or other HR gene defects (e.g. PALB2, RAD51C/D, ATM) lack efficient DSB repair, making them highly sensitive to PARP inhibition ⁸ ⁷. Conversely, *upregulation* or restoration of HR (via somatic **BRCA reversion mutations** or loss of end-joining factors like 53BP1/Shieldin) confers *resistance* ¹² ¹⁷. Indeed, tumors that acquire second-site mutations restoring BRCA1/2, or lose 53BP1 (TP53BP1) and related

factors (RIF1, REV7, shieldin) to resume DNA end resection, regain HR capacity and resist talazoparib ¹² ¹⁷ .

- **Apoptotic signaling [R-HSA-109581]:** Enhanced pro-apoptotic signaling drives talazoparib sensitivity, whereas defects in apoptosis promote resistance. Talazoparib-induced DNA damage ultimately triggers apoptosis in HR-deficient cells ⁹ ¹¹ . Therefore, upregulation of apoptotic pathways (e.g. pro-death BCL-2 family effectors) would amplify talazoparib's effect. By contrast, *downregulation* of apoptosis (e.g. overexpression of anti-apoptotic BCL-2 or loss of p53 function) can blunt cell death and lead to resistance. For example, intact p53-dependent checkpoints and apoptosis are required for full PARPi response ⁹ .
- **Drug Efflux and Metabolism [R-HSA-382556]:** Overexpression of ATP-binding cassette (ABC) transporters (especially **ABCB1/MDR1/P-gp**) exports talazoparib out of cancer cells, causing *resistance*. Experimental models have shown that upregulation or gene amplification of ABCB1 markedly reduces intracellular PARPi levels and confers PARPi resistance ¹⁸ . Conversely, loss or inhibition of efflux pumps sensitizes cells. Clinically, ABCB1 overexpression has been implicated in resistance to olaparib/talazoparib ¹⁸ .
- **DNA Damage Checkpoints (ATR/ATM) [R-HSA-176187]:** The ATR/CHK1 and ATM checkpoint pathways sense replication stress and coordinate repair. *Upregulation* of ATR/ATM-mediated checkpoints provides cancer cells more time to repair PARP-induced damage, contributing to resistance ¹⁹ . In fact, concurrent ATR activation promotes HR restoration and survival under PARP inhibition ¹⁹ . Inhibition of ATR or ATM kinases synergizes with PARP inhibitors in preclinical models, underscoring that **downregulating** these checkpoints increases sensitivity. Thus, activation of the ATR/ATM pathway opposes talazoparib efficacy ¹⁹ .
- **PI3K/AKT Pro-survival Signaling [R-HSA-198203]:** Activation of the PI3K/AKT/mTOR pathway confers **resistance** by promoting cell survival and DNA repair. PARP inhibition can paradoxically activate PI3K/AKT signaling ²⁰ . Upregulated AKT signaling enhances HR and inhibits apoptosis, diminishing PARPi lethality ²⁰ . By contrast, *inhibition* of PI3K/AKT (e.g. via PTEN expression) can sensitize tumors to talazoparib ²⁰ ²¹ . Indeed, PTEN loss (which activates AKT) has complex effects: it can induce HR deficiency and PARPi sensitivity ²¹ , but in some contexts fosters PARPi resistance via PI3K/AKT ²¹ ²⁰ .
- **Replication Fork Stability:** Stabilization of stalled replication forks is a resistance mechanism. Normally, BRCA1/2 protect forks from degradation; cancer cells can bypass this by downregulating nucleases (MRE11, MUS81) or fork remodeling factors, effectively shielding forks even without BRCA ²² ²³ . Such *fork protection* allows cell survival under PARPi despite HR deficiency, leading to resistance ²² ²³ .

Collectively, talazoparib sensitivity in breast cancer correlates with **HRD genotype** and active apoptotic responses, whereas resistance associates with **HR restoration, enhanced DNA damage checkpoints or survival signaling, and drug efflux** ¹² ¹⁹ . These insights guide combination strategies (e.g. ATR or PI3K inhibitors) to overcome PARPi resistance.

Summary of Pathway Effects

Pathway (ID/Name)	Regulation ↑ / ↓	Talazoparib Effect	Biological Rationale	Reference
Homologous Recombination (R-HSA-5685942)	↓ (down)	Sensitive	HR deficiency (e.g. BRCA1/2 loss) prevents DSB repair, causing PARP-trapped lesions and cell death ⁸ . Synthetic lethality occurs when HR is lost.	⁸ ⁷
(HDR through HR)	↑ (up)	Resistant	Restoration of HR (via BRCA reversion or loss of 53BP1–shieldin) repairs DSBs, abrogating PARPi lethality ¹² ¹⁷ .	¹² ¹⁷
Apoptosis (R-HSA-109581)	↑ (up)	Sensitive	Talazoparib-induced DNA damage triggers apoptosis; enhanced apoptotic signaling promotes tumor cell death ⁹ ¹¹ .	⁹ ¹¹
(Programmed cell death)	↓ (down)	Resistant	Impaired apoptosis (e.g. overexpressed BCL-2 or p53 loss) allows survival despite DNA damage, reducing PARPi efficacy ⁹ .	⁹ ¹¹
ABC Transporters (R-HSA-382556)	↑ (up)	Resistant	Overexpression of ABCB1/P-glycoprotein pumps exports talazoparib out of cells, lowering intracellular drug and conferring PARPi resistance ¹⁸ .	¹⁸
	↓ (down)	Sensitive	Reduced efflux increases intracellular talazoparib, enhancing cytotoxicity. (Hypothetical inverse of ↑)	¹⁸
ATR-mediated checkpoint (R-HSA-176187)	↑ (up)	Resistant	ATR/ATM checkpoint activation slows cell cycle and promotes repair; upregulated ATR facilitates HR restoration and survival under PARP inhibition ¹⁹ .	¹⁹
	↓ (down)	Sensitive	Inhibiting ATR/ATM prevents damage repair, exacerbating replication stress and increasing talazoparib cytotoxicity ¹⁹ .	¹⁹

Pathway (ID/Name)	Regulation ↑ / ↓	Talazoparib Effect	Biological Rationale	Reference
PI3K/AKT signaling (R-HSA-198203)	↑ (up)	Resistant	PI3K/AKT activation promotes cell survival and HR repair; upregulation (e.g. via PTEN loss) counteracts PARPi-induced apoptosis ²⁰ .	²⁰
	↓ (down)	Sensitive	Inhibiting PI3K/AKT lowers survival signaling and HR capacity, sensitizing cells to PARP inhibition ²⁰ ²¹ .	²⁰ ²¹

Each pathway above is based on human breast cancer data: for example, the *alternate repair* and *efflux* mechanisms have been identified in human tumor analyses and cell lines ¹² ¹⁸. The synthetic lethality rationale is firmly established in BRCA-mutant breast cancer ⁷ ⁸. Notably, the evidence indicates that BRCA1 and BRCA2 carriers (basal-like/TNBC and luminal subtypes, respectively) respond similarly to talazoparib ¹⁶, and that additional common mutations (e.g. *TP53*, *PIK3CA*) do not by themselves negate sensitivity ¹⁶.

References: Authoritative sources including DrugBank ⁷, Reactome/MSigDB pathways, and peer-reviewed studies of PARP biology ⁸ ¹² ²⁰ were used to compile this report.

¹ ² ³ ⁴ ⁶ ⁷ ⁹ ¹³ ¹⁵ Talazoparib: Uses, Interactions, Mechanism of Action | DrugBank Online
<https://go.drugbank.com/drugs/DB11760>

⁵ Germline BRCA-Mutated HER2-Negative Advanced Breast Cancer: Overcoming Challenges in Genetic Testing and Clinical Considerations When Using Talazoparib - PubMed
<https://pubmed.ncbi.nlm.nih.gov/37246120/>

⁸ ¹¹ ¹² ¹⁷ ¹⁸ ¹⁹ ²⁰ ²² ²³ Alternate therapeutic pathways for PARP inhibitors and potential mechanisms of resistance | Experimental & Molecular Medicine
https://www.nature.com/articles/s12276-021-00557-3?error=cookies_not_supported&code=10408e90-9b6d-4603-b46a-28d64f42df46

¹⁰ Talazoparib | Pfizer Oncology Development Website
<https://www.pfizeroncologydevelopment.com/molecule/talazoparib>

¹⁴ Talazoparib versus chemotherapy in patients with germline BRCA1/2-mutated HER2-negative advanced breast cancer: final overall survival results from the EMBRACA trial - PubMed
<https://pubmed.ncbi.nlm.nih.gov/32828825/>

¹⁶ Genomic analysis of advanced breast cancer tumors from talazoparib-treated gBRCA1/2mut carriers in the ABRAZO study | npj Breast Cancer
https://www.nature.com/articles/s41523-023-00561-y?error=cookies_not_supported&code=75753832-6a95-4f0a-97a3-97c2240482c5

²¹ Phase II study of talazoparib in advanced cancers with BRCA1/2, DNA repair, and PTEN alterations | npj Precision Oncology

https://www.nature.com/articles/s41698-024-00634-6?error=cookies_not_supported&code=0c16203a-95db-43ef-a436-3422f3bf2751